

THEME 07: BRAIN, NEURAL AND REHABILITATION ENGINEERING

WORKSHOP

Exploring Altered States of Consciousness Through EEG and Brain-Computer Interfaces

Sébastien Rimbert, Laurent Bougrain, Jérémie Mattout

clouding of consciousness

confusional state

lethargy

delirium

obtundation

stupor

hypersomnia

vegetative state

dementia

akinetic mutism

coma

anesthesia

brain death

locked-in syndrome

COPENHAGEN

**IEEE
EMBS**
IEEE Engineering
Medicine & Biology Society

47TH ANNUAL
International Conference of the
**IEEE Engineering in Medicine
and Biology Society**

Altered level of consciousness defined from reduction in response to stimuli

Grady Coma Scale

(grades I to V)

Grade	State of awareness	Responds appropriately to:		
		Calling name	Light pain	Deep pain
I	Confused, drowsy, lethargic, indifferent and/or uncooperative; does not lapse into sleep when left undisturbed	Yes	Yes	Yes
II	Stuporous; may be disoriented to time, place, and person; will lapse into sleep when not disturbed; or belligerent and uncooperative	No	Yes	Yes
III	Deep stupor; requires strong pain to evoke movement	No	No	Yes
IV	Exhibits decorticate or decerebrate posturing to a deep pain stimulus	No	No	No
V	Does not respond to any stimuli; flaccid	No	No	No

Tindall SC. Level of Consciousness. Chap. Clinical Methods: The History, Physical, and Laboratory Examinations, 1990]

Glasgow Coma Scale

(score ranges from 15 to 3)

Eye opening (E)

4. Opens eyes spontaneously
3. Opens eyes to voice
2. Opens eyes to pain
1. No eye opening

Best motor response (M)

6. Obeys commands
5. Localizes to pain
4. Withdraws to pain
3. Abnormal flexor response
2. Abnormal extensor response
1. No movement

Best verbal response (V)

5. Appropriate and oriented
4. Confused conversation
3. Inappropriate words
2. Incomprehensible sounds
1. No sounds

[Teasdale G, Jennett B. Assessment of coma and impaired consciousness. A practical scale. The Lancet, 1974]

AVPU

(scale from A to U)

Alert:

fully awake
spontaneously open eyes
respond to voice
bodily motor function

Verbal:

eyes, voice or motor
response to voice

Pain:

Voice or motion response
to pain stimulus (sternal
rub, squeezing the
fingers,...)

Unresponsive:

no eye, voice or motor
response to voice or
pain.

[McNarry, AF; Bateman, DN . Simple bedside assessment of level of consciousness: comparison of two simple assessment scales with the Glasgow Coma. Anaesthesia, 2004]

ACDU

(scale from A to U)

Alertness

Confusion

Drowsiness

Unresponsiveness

Altered level of consciousness to diagnose disorders, diseases and causes

Most common causes of coma:

- metabolic and toxic disturbances of the brain such as acid–base disequilibrium,
- disorders of oxygen or glucose metabolism,
- uremic and hepatic encephalopathy,
- drug overdose,
- poison ingestion.

Stupor or coma

- Cerebrovascular disorders such as ischemic or embolic stroke,
- intracerebral or subarachnoid hemorrhage,
- Infection (meningitis, cerebral abscess, or encephalitis).

Altered consciousness:

- epilepsy and various postconvulsive states,
- Intracranial sequelae of head,
- brain tumors,
- primary or metastatic.

Workshop Program

- ***General workshop introduction***

Laurent Bougrain (Université de Lorraine, CNRS, LORIA, Nancy, France / Sorbonne Université, Institut du Cerveau – Paris Brain Institute - ICM, CNRS, Inria, Inserm, AP-HP, Hôpital de la Pitié Salpêtrière, 75013 Paris, France)

- ***Preserved Neural Functions and Outcome Prediction on the First Days of Coma***

Marzia De Lucia (Lausanne University Hospital & University of Lausanne)

- ***To What Extent can Cognitive and Motor Skills be Dissociated in DOC Patients?***

Jérémy Mattout (Centre de Recherche en Neurosciences de Lyon - Inserm U1028 / CNRS UMR5292, 69675 BRON, France)

- ***Detecting Accidental Awareness during General Anesthesia with Neuromuscular Blockade: Challenges and Considerations***

Sebastian Halder (University of Essex, UK)

break

- ***Tracking Consciousness with Median Nerve Stimulation-Based BCI During General Anesthesia***

Valérie Marissens-Cueva (Inria Center at the University of Bordeaux / LaBRI, France and Université de Lorraine, CNRS, LORIA, Nancy, France)

- ***From Movement to Consciousness & vice-versa***

Guy Chéron (Université libre de Bruxelles)

- ***Summary & Discussion***

Sébastien Rimbart (Inria Center at the University of Bordeaux / LaBRI, France)